Project Gravitas Derivations

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1 Derivation of orbital velocity

From Newton's law of universal Gravitation:

$$F_g = \frac{m_1 m_2 G}{r^2}$$

We make the substitution $F_g = \frac{m_1 v^2}{r}$ since gravity acts as the centripetal force (we are assuming perfectly circular orbits, or $e = 0$)

$$\frac{m_1 v^2}{r} = \frac{m_1 m_2 G}{r^2}$$

$$\frac{v^2}{r} = \frac{m_2 G}{r^2}$$

$$v^2 = \frac{m_2 G r}{r^2}$$

$$v^2 = \frac{m_2 G}{r}$$

$$v = \sqrt{\frac{m_2 G}{r}}$$

Here, v is the tangential velocity.

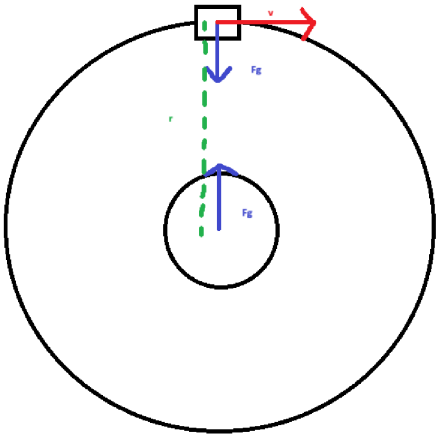


Figure 1: A diagram which the equation above represents

2 Derivation of orbital period

From the definition of speed:

$$s = \frac{d}{t}$$

we rearrange this for t:

$$t = \frac{d}{s}$$

We can now plug in $d = 2\pi r$ (because the object is moving in a circular orbit of radius r and $s = v = \sqrt{\frac{m_2 G}{r}}$ (from our derivation above) to yield:

$$T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{m_2 G}{r}}}$$

3 Derivation of acceleration due to gravity

We start off with Newton's Law of Universal Gravitation:

$$F_g = \frac{m_1 m_2 G}{r^2}$$

We substitute $F_g = m_1 g$ where g is the acceleration due to gravity

$$m_1 g = \frac{m_1 m_2 G}{r^2}$$

$$g = \frac{m_2 G}{r^2}$$

Notice how the mass of the smaller object that is orbiting around the larger one does not play a role in orbital velocity, orbital period, or acceleration due to gravity!